

Figure 6: An algebraic function as surf

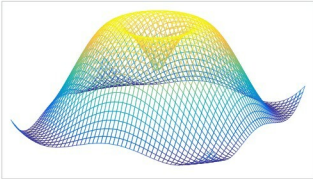


Figure 7: An algebraic function as mesh

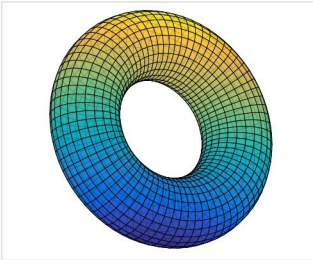


Figure 8: A parametric torus

```
\addplot3[
    surf,
    domain=-230:230,
    samples=50,
    opacity=0.8,
    shader=flat,
    line width=0.1pt
]
{sin(sqrt(x^2+y^2))};
\end{axis}
```

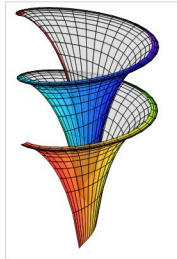


Figure 9: The Dini's surface

```
\end{tikzpicture}
\end{document}
```

An alternative is to replace *surf* with *mesh*. The final plots are shown in Figures 6 and 7. If the surface is parametric, for example, a torus (Figure 8), the three equations can be written in an easy way putting them between '(' and ')':

```
((cos(x)),
{(2.5+sin(x))*cos(y)},
{(2.5+sin(x))*sin(y)});
```

Plotting a 3D function, algebraic or parametric, requires the customisation of some keys — for example, *domain* (x range, or x, y if *domain* y is not specified), *domain y* (y range), *draw* (lines colour), *line width* (lines width), *opacity* (alpha value, from 0 - totally transparent - to 1 - totally opaque -), *samples* (which manages the number of sampled points on x - its default value is 25 -, or x, y if *samples* y is not specified), *samples y* (manages the number of sampled points on y), *shader* (lines shade), and *z buffer* (if equal to *sort*, the segments more distant from the observation point are first plotted).

I think that a better way to plot a parametric function is shown in Reference 7 or Reference 14 (page 109), in which the package *pst-solides3d* is used. This is the only example discussed here for which it's necessary to run with XeLaTeX. The final plot in Figure 9 shows the Dini's surface (from Ulysse Dini, http://en.wikipedia.org/wiki/Ulysse_Dini). Some important options are: *viewpoint* (notation with spherical coordinates), *decran* (a particular distance, see Reference 14, pages 14-15), *base* (dimensions of the grid), and *ngrid* (the number of grid lines).

```
\documentclass[border=0.5cm]{standalone}
\usepackage{pst-solides3d}
\begin{document}
\begin{pspicture}(-1,-2)(1,2)
\psset(viewpoint=20 70 20 rtp2xyz,Decran=20)
\defFunction[algebraic]{lily}(u,v)
{
    {cos(u)*sin(v)}
    {sin(u)*sin(v)}
    {cos(v)+ln(tan(v/2))+0.15*u}
}
\psSolid[
    object=surfaceparametree,
    function=lily,
    base=0 12.56 0.13 2,
    ngrid=60 0.1,
    linewidth=0.1pt,
```